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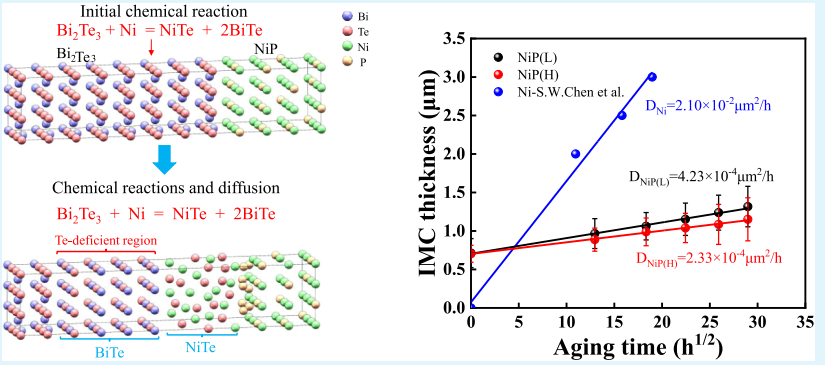
Extremely Low Contact Resistivity of Bi_2Te_3-Based Modules Enabled by NiP-Based Alloy Barrier

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ABSTRACT: Electrode diffusion barrier plays an important role in thermoelectric cooling devices. Compared with p-type $Bi_{0.5}Sb_{1.5}Te_3$, the compatibility between commercial Ni barrier and n-type $Bi_2Te_{2.7}Se_{0.3}$ is a key bottleneck to enhance the performance of Bi_2Te_3-based cooling devices. This paper proposed a NiP alloy barrier to improve the compatibility with n-type $Bi_2Te_{2.7}Se_{0.3}$, and systemically investigated the contact and interfacial dynamics properties. Due to the low diffusion rate of NiP alloy, the initial interfacial contact resistivity of $Bi_2Te_{2.7}Se_{0.3}$/NiP is as low as 0.90 $\mu\Omega$ cm^2, and it further can be depressed below 1.98 $\mu\Omega$ cm^2 even after aging at 423 K for 35 days, indicating the superior thermal stability of the NiP barrier layer compared to the commercial Ni barrier layer. Based on the NiP barrier, a 15-pair bismuth telluride device is prepared and a high cooling temperature difference of 71.5 K at a hot-side temperature of 304 K is achieved, which proves the practical applications potential of NiP barrier for Bi_2Te_3-based modules.

KEYWORDS: *contact resistance, electroless NiP, thermal stability, IMC, Bi_2Te_3devices*

1. INTRODUCTION

Thermoelectric (TE) conversion technology based on semiconductor materials is an environmentally friendly technology that enables the direct interconversion of thermal and electrical energy. TE devices consist of ceramic substrates, electrodes, metallization layers, and thermoelectric materials and are packaged by solder in thermal parallel and electrical series. Its advantages such as no mechanical moving parts, no noise, zero greenhouse gas emissions, excellent stability, high reliability, compact structure, fast response, etc. have attracted widespread attention.[1,2] With the rapid development of TE conversion technology, its application has been successfully integrated into human production life, such as the automotive industry,[3,4] industrial waste heat recycling,[5,6] deep space exploration,[7−9] medical and health care,[10−12] consumer electronics,[13−15] and other fields of application.

The output performance of TE devices depends not only on the material figure of merit (ZT, $ZT = S^2\sigma T/\kappa$, where S, σ, κ, and T are the Seebeck coefficient, electrical conductivity, thermal conductivity, and absolute temperature, respectively) but is also affected by the devices' welding quality and structure design. Considering the electrode contact resistance, the effective $\langle ZT\rangle_D$ of TE device can be expressed as[16,17]

$$\langle ZT\rangle_D = \frac{L}{(L + 2R_c\sigma)}\langle ZT\rangle_M$$

where L is the length of the thermoelectric leg, R_c is the contact resistance between the electrode and thermoelectric materials, and $\langle ZT\rangle_M$ is the effective ZT of the thermoelectric

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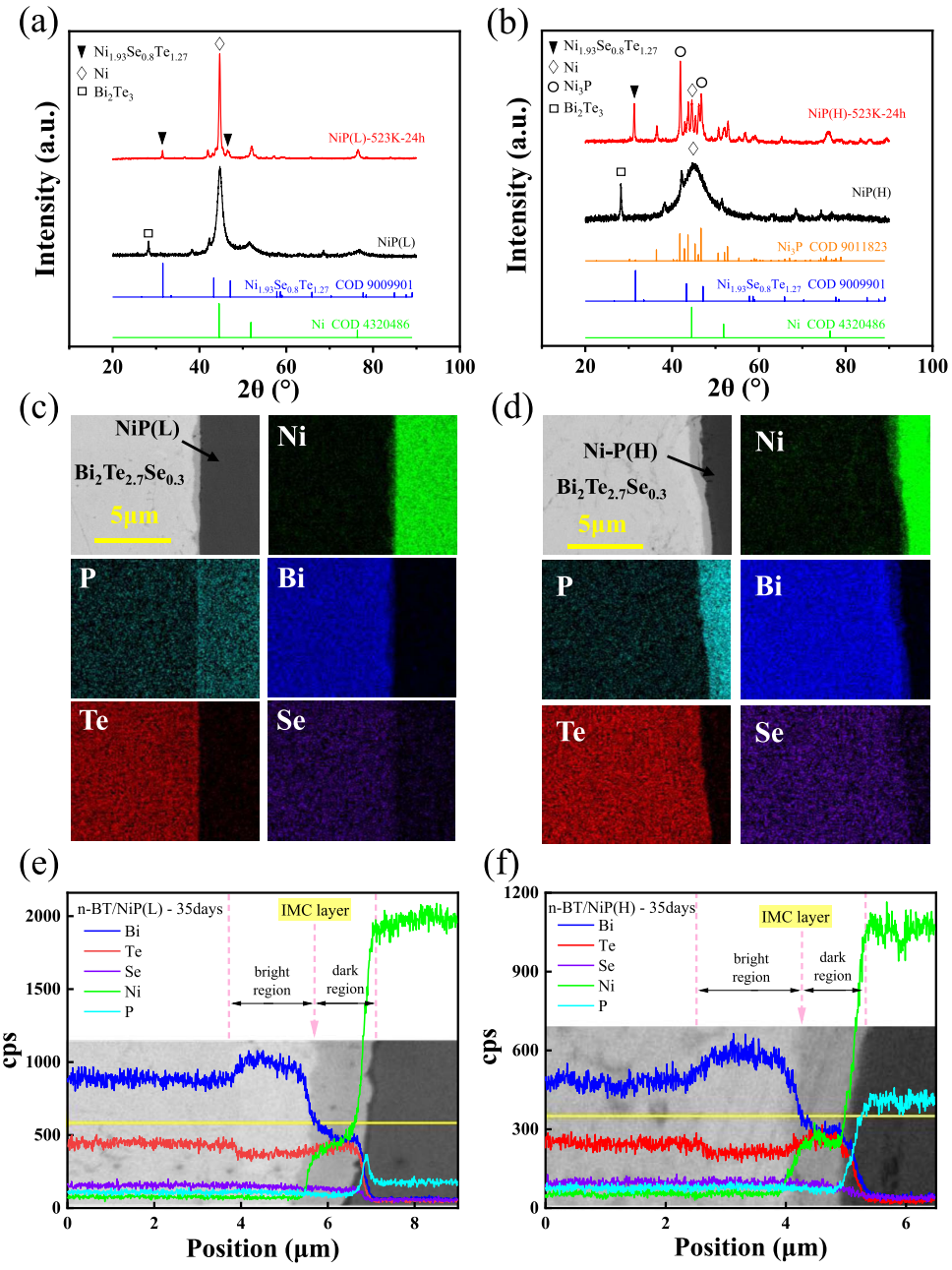


Figure 1. XRD patterns of initial and aging at 523 K for 24 h: (a) $Bi_2Te_{2.7}Se_{0.3}$/NiP(L), (b) $Bi_2Te_{2.7}Se_{0.3}$/NiP(H). Initial SEM image and elemental mapping of (c) $Bi_2Te_{2.7}Se_{0.3}$/NiP(L) and (d) $Bi_2Te_{2.7}Se_{0.3}$/NiP(H). SEM-EDS line scan of elemental changes aged at 423 K for 35 days of (e) $Bi_2Te_{2.7}Se_{0.3}$/NiP(L) and (f) $Bi_2Te_{2.7}Se_{0.3}$/NiP(H).

material between the hot side (T_h) and cold side (T_c). For a typical device with a thermoelectric leg of $L \sim 1$ mm and $\sigma \sim 10^5$ S m^{-1}, R_c should be much less than 10 $\mu\Omega$ cm^2 for the high performance of TE devices. Consequently, concerted efforts should be directed toward minimizing the influence of R_c on the device's efficiency.

Bi_2Te_3-based device packages are usually soldered with Sn or Sn-based solders, and Te can rapidly diffuse into the Sn-based solder at the soldering temperature, forming a brittle and porous Sn−Te intermetallic compound (IMC), which results in increased contact resistance and thermal resistance, a weak mechanical joint strength, and even leading to device soldering failure.[18−21] Therefore, it is vital to inhibit the formation of the Sn−Te IMC layer. Numerous studies have shown that Ni as a barrier layer can effectively block the diffusion of elements between the solder and Bi_2Te_3; moreover, its welding wettability is better than that of Bi_2Te_3.[22−24] Liu et al. used one-step hot press sintering of Ni barrier layer to obtain p-type $Bi_{0.4}Sb_{1.6}Te_3$ initial contact resistivity <1 $\mu\Omega$ cm^2, but in n-type $Bi_2Te_{2.7}Se_{0.3}$, the formation of p-type region $(Bi_{1-x}Ni_x)_2(Te,Se)_{3-y}$ at the interface resulted in an initial interfacial contact resistivity of 210 $\mu\Omega$ cm^2.[17] Tang et al. obtained initial contact resistivities of 5.31 and 3.64 $\mu\Omega$ cm^2 with Ni as the barrier layer for n-type and p-type Bi_2Te_3, respectively, while aging at 423 K for 42 days, cracks appeared at the interfacial junction between the n-type Bi_2Te_3.[25] Joshi et al. adopted pulsed light annealing prior to nickel metallization for recrystallization of the surface of Bi_2Te_3 thermoelectric arms, producing the contact resistivity of 7.7 and 5.2 $\mu\Omega$ cm^2 for n-type and p-type, respectively, as well as the connection

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Table 1. IMC Composition at the n-BT/NiP Interface

NiP	annealing duration (day)	composition of dark region (atom %)				composition of bright region (atom %)		
		Te	Bi	Se	Ni	Te	Bi	Se
NiP(L)	7	38.13	13.41	1.44	47.03	48.10	48.99	2.92
	14	35.11	13.16	1.85	49.88	49.54	47.58	2.87
	21	36.38	14.77	2.10	46.74	46.75	50.54	2.72
	28	43.06	13.91	1.28	41.76	48.62	48.42	2.96
	35	39.68	17.68	2.31	40.32	47.09	48.88	4.03
NiP(H)	7	38.37	15.41	2.52	43.97	47.99	49.04	2.97
	14	38.12	14.13	2.05	45.71	47.32	49.16	3.52
	21	39.00	14.64	2.32	44.04	48.83	48.66	2.61
	28	40.91	14.96	2.32	41.84	48.08	48.80	3.12
	35	41.34	15.57	1.53	41.56	47.94	48.79	3.27

strength of 8 MPa.[26] Despite the advantages of the Ni barrier layer to reduce the R_c, especially for p-type $Bi_{0.5}Sb_{1.5}Te_3$, n-type $Bi_2Te_{2.7}Se_{0.3}$ (n-BT) has a larger interfacial contact resistivity and poorer stability, which needs to be further optimized.

Recently, the high entropy effect led by multicomponent alloying arouses great attention in the fields of thermoelectric, hard alloys, and other materials.[27] Multialloy barrier layers can be very good for minimizing lattice mismatch, thus elevating the diffusion barrier energy and suppressing the diffusion across the bulk lattice.[28,29] For the Bi_2Te_3-based devices, constructing multicomponent alloy barriers based on Ni is expected to highly improve the interfacial performance, such as Ni−P barrier layers with high bond strength and solder wettability that have been successfully used in commercial electronics packaging.[30,31] In this study, NiP barrier layers with different phosphorus content (low phosphorus-NiP(L), high phosphorus-NiP(H)) were prepared to form an n-BT/NiP/Cu/NiP/n-BT sandwich structure for the investigation of R_c. The contact resistance between NiP and n-BT, NiP and solder, and solder and Cu electrode was accurately measured using a home-built micron-level automatic scanning probe, and the reliability of the sample soldering was verified by aging for different times. n-BT/NiP(L) has an initial low interfacial contact resistivity of 0.90 $\mu\Omega$ cm^2, which is slightly increased (1.98 $\mu\Omega$ cm^2) after it was aged at 423 K for 35 days. Such excellent interfacial contact resulted in a maximum cooling temperature difference (ΔT_{max}) of 71.5 K of thermoelectric cooling devices (TEC) at the hot-side temperature of 304 K, and the TEC demonstrates a high shear strength of ∼12.5 MPa.

2. EXPERIMENTAL PROCEDURES

2.1. Sample Preparation of NiP Barrier Layer. The commercially extruded n-type $Bi_2Te_{2.7}Se_{0.3}$ rods are first sliced into thin slices with 1 mm in thickness. Prior to the metallization of the n-type $Bi_2Te_{2.7}Se_{0.3}$, the sample surface undergoes mechanical sand-blasting using Al_2O_3 powder. This process ensures a uniform and rough surface for further processing. The surface roughness of the n-BT bulk is measured using a laser confocal scanning microscope (VK-X1100, Japan). To prepare the samples for electroless plating with NiP barrier layer, they are subjected to ultrasonic cleaning with deionized water for 3 min followed by acetone for 10 min. This cleaning step removes surface impurities and oxides, ensuring a clean surface for plating. Next, using Guangzhou Yi Shun Chemical Co. Ltd., China (Q/YS.602-low phosphorus content) and (Q/YS.602-high phosphorus content) chemical plating baths, the NiP alloy barrier layer with 5 and 20% phosphorus was obtained by placing the wafers into the plating bath at 85 °C for 10 times, respectively, resulting in the deposition of a 3−5 μm NiP barrier layer on the sample surface.

This sequence of steps ensures the proper preparation of the n-BT samples for subsequent analysis and testing, providing a reliable and consistent NiP barrier layer on the sample surface. Soldering of two n-BT with NiP barrier layer (Φ30 × 1.0 mm^3) to copper foil (Φ30 × 0.08 mm^3) with $Sn_{95}Sb_5$ (melting point of 513 K) was carried out with a solder. A scribing machine (NANO150G, Taiwan, China) was employed to cut the disk sample into 2.3 × 2.3 mm^2. The samples were vacuum-sealed into quartz tubes and aged at 423 K for 7, 14, 21, 28, and 35 days. The contact resistance between the NiP barrier layer and n-BT was tested using a home-built micrometer-level interface resistance test system. The phase composition of NiP/$Bi_2Te_{2.7}Se_{0.3}$ before and after aging using X-ray diffraction (XRD) equipped with Cu Kα radiation of 1.5406 Å at room temperature. The IMC morphology and chemical composition were examined by scanning electron microscopy (SEM, Thermo Scientific Apreo 2 S HiVac, 10 kV) equipped with an energy-dispersive spectrometer (EDS). A bond tester (DAGE4000) was used to measure the TEC shear strength of the connection at the interface layer.

2.2. Fabrication of TEC and Performance Measurements. The n-type and p-type Bi_2Te_3 wafers with NiP(L) barrier layers were cut into 1 × 1 × 1.6 mm^3 thermoelectric legs using a scribing machine, respectively. $Sn_{95}Sb_5$ solder paste was screen-printed on the upper and lower copper-clad substrates, and the n and p thermoelectric legs were placed in electric series on the solder joints and then encapsulated and solder. Digital bridge was used to measure the internal resistance of the thermoelectric components. And maximum temperature difference of TEC was tested.

3. RESULTS AND DISCUSSION

3.1. Interfacial Dynamical Analysis of $Bi_2Te_{2.7}Se_{0.3}$/NiP. In order to investigate the stability of NiP barrier layers with $Bi_2Te_{2.7}Se_{0.3}$, XRD analysis was performed on NiP barrier layers after chemical plating and aged at 523 K for 24 h. As shown in Figure 1a,b, initially, diffraction peaks of Ni were observed for both NiP(L) and NiP(H) layers, and it shows more amorphous behavior for NiP(H) layers since the main peak of Ni in NiP(H) layers is rather broadened. After being aged at 523 K for 24 h, diffraction peaks of $Ni_{1.93}Se_{0.8}Te_{1.27}$ were present in NiP(L)/$Bi_2Te_{2.7}Se_{0.3}$, but the resultants of NiP(H)/$Bi_2Te_{2.7}Se_{0.3}$ are more complicated, including Ni_3P and $Ni_{1.93}Se_{0.8}Te_{1.27}$ at least. Figure 1c,d shows the SEM element maps of $Bi_2Te_{2.7}Se_{0.3}$ with NiP(L) and NiP(H) interfaces aged at 423 K for 35 days.

Due to the presence of a concentration gradient, which allows interdiffusion of elements near the interface at high temperatures, there are interfacial reaction layers consisting of dark and bright regions between n-BT and NiP(L). Figure 1e,f shows SEM-EDS line scans of elemental changes in $Bi_2Te_{2.7}Se_{0.3}$ with NiP(L) and NiP(H) aged at 423 K for 35

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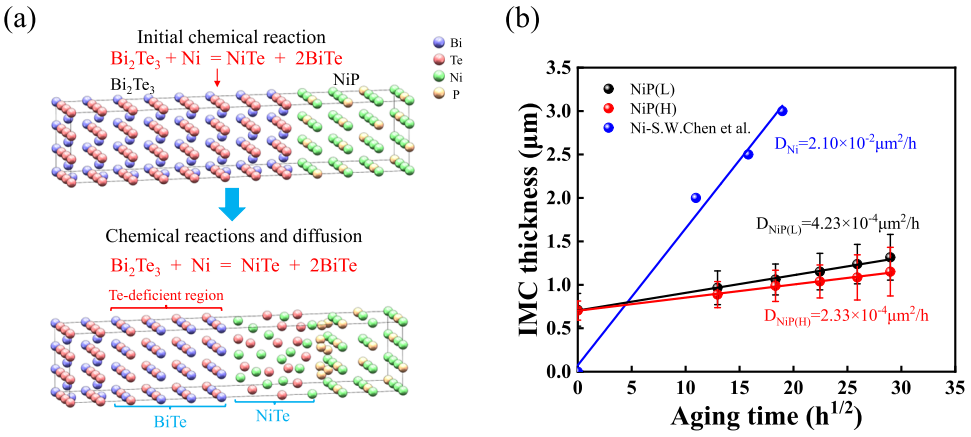


Figure 2. (a) Reaction and diffusion model of Bi_2Te_3/NiP joint, (b) $Bi_2Te_{2.7}Se_{0.3}$ substrate with Ni,[32] NiP diffusion rates of dark region at 423 K.

days. It can be observed that before the IMC layer in the dark region, there is an area of variation in the elemental content of Te and Bi, which is consistent with the previous reports.[24,25] The Te content decreases first and then reaches the peak in IMC, even higher than the Te content in $Bi_2Te_{2.7}Se_{0.3}$. Additionally, a Bi-rich region is observed in the Te-deficient region, with the Bi content being much higher than in the $Bi_2Te_{2.7}Se_{0.3}$. As the aging time prolongs, the Te-deficient region in $Bi_2Te_{2.7}Se_{0.3}$ is expanding, and similarly, the Bi-rich region has also expanded correspondingly. Table 1 lists the elemental content of IMC layers for NiP(L) and NiP(H) barrier layers with substrate aged at 423 K for 7, 14, 21, 28, and 35 days. For the diffusion of elements in the NiP(L) barrier layer, the Ni content decreases sharply at the IMC, and there is a P-rich region at the edge of the barrier layer, but the P element is virtually undetected in the IMC layer. This indicates that the diffusion rate of P is much lower than that of Ni. These results demonstrated that the NiP barrier layers with different phosphorus contents are able to successfully block the diffusion of Sn-based compounds from the solder to the n-BT bulk and that of Te, Bi, and Se elements from the n-BT bulk to the solder. The formation of IMC at the interface enhances the mechanical strength of the joint. However, excessive growth of IMC compounds and overdiffusion of elements would increase brittleness, reduce thermal stability, and degrade electrical properties. Therefore, it is important to control the interfacial reactions and diffusion processes to ensure the integrity and reliability of the joint.

From the compositions of the bright region with different aging times in Table 1, it can be found that the bright region is mainly BiTe phase. And based on the analysis of the content of elements, the dark region can be identified as NiTe phase, which is consistent with the presence of $Ni_{1.93}Se_{0.8}Te_{1.27}$ phase shown in the XRD pattern of NiP(L)/$Bi_2Te_{2.7}Se_{0.3}$ (Figure 1a). Since both Se and phosphorus have very low contents in the substrate $Bi_2Te_{2.7}Se_{0.3}$ bulk and NiP(L) barrier layer, the reaction at the $Bi_2Te_{2.7}Se_{0.3}$/NiP(L) joint can be determined as $Bi_2Te_3 + Ni \rightarrow NiTe + BiTe$.

The interfacial reaction produced NiTe compounds in the dark region, and due to the existence of the elemental concentration gradient at the interface, the diffusion and consumption of chalcogen elements led to the appearance of BiTe compounds within the substrate in the bright region. Figure 2a shows a schematic of the reaction and diffusion models at the n-BT/NiP interface. With the chemical reaction and elemental diffusion, more Ni and Te are generated during the chemical reaction at the interface, and the elemental diffusion leads to further extension of the Te-deficient region, i.e., the diffusion leads to a larger bright region of BiTe. As a consequence of the reciprocal diffusion of Ni and Te elements between the NiP barrier layer and the substrate, a distinct dark region of IMC is generated. In addition, experimental observations reveal P enrichment near the NiP(L) barrier layer, while this phenomenon does not appear in NiP(H). The possible reason for this phenomenon is that in NiP(L) barrier layer, P exists in a free state, whereas in NiP(H), a Ni_3P alloy compound is formed (Figure 1b) due to the higher P content, which prevents the diffusion of P elements.

The thicknesses of the NiTe interfacial reaction layer in the dark region at the n-BT/NiP interface were measured by SEM and are shown in Table 2. However, due to the fact that the compositions of BiTe and $Bi_2Te_{2.7}Se_{0.3}$ substrates are very close, the thickness of the bright region is hard to precisely determine by SEM. According to the conservation of total elements, the reactions occurring at the interface can be given: $Bi_2Te_3 + Ni = NiTe + 2BiTe$. The thickness of NiTe in the dark region is easily measured, and the thickness of the BiTe layer in the bright region can be calculated from the molar ratio.

Table 2. Thickness of the NiTe Interfacial Reaction Layer in the Dark Region at the n-BT/NiP Interface

annealing duration (day)	n-BT/NiP(L) (μm)	n-BT/NiP(H) (μm)
0	0.71 ± 0.09	0.70 ± 0.06
7	0.97 ± 0.10	0.89 ± 0.08
14	1.06 ± 0.09	0.99 ± 0.09
21	1.15 ± 0.11	1.04 ± 0.10
28	1.24 ± 0.12	1.09 ± 0.13
35	1.32 ± 0.13	1.15 ± 0.14

$$L_{BiTe} = 2Y_{NiTe}\cdot(M_{Bi} + M_{Te})/(S\cdot\rho_{BiTe}) = 2\rho_{NiTe}\cdot S\cdot L_{NiTe}/(M_{Ni} + M_{Te})\cdot(M_{Bi} + M_{Te})/(S\cdot\rho_{BiTe})$$

where Y_{NiTe} denotes the generated number of moles of NiTe, M_{Bi}, M_{Te}, and M_{Ni} denote the atomic weights of Bi, Te, and Ni, respectively; ρ_{BiTe} = 8.25 g/cm^3 and ρ_{NiTe} = 8.38 g/cm^3 denote the theoretical densities of BiTe and NiTe, respectively; S is

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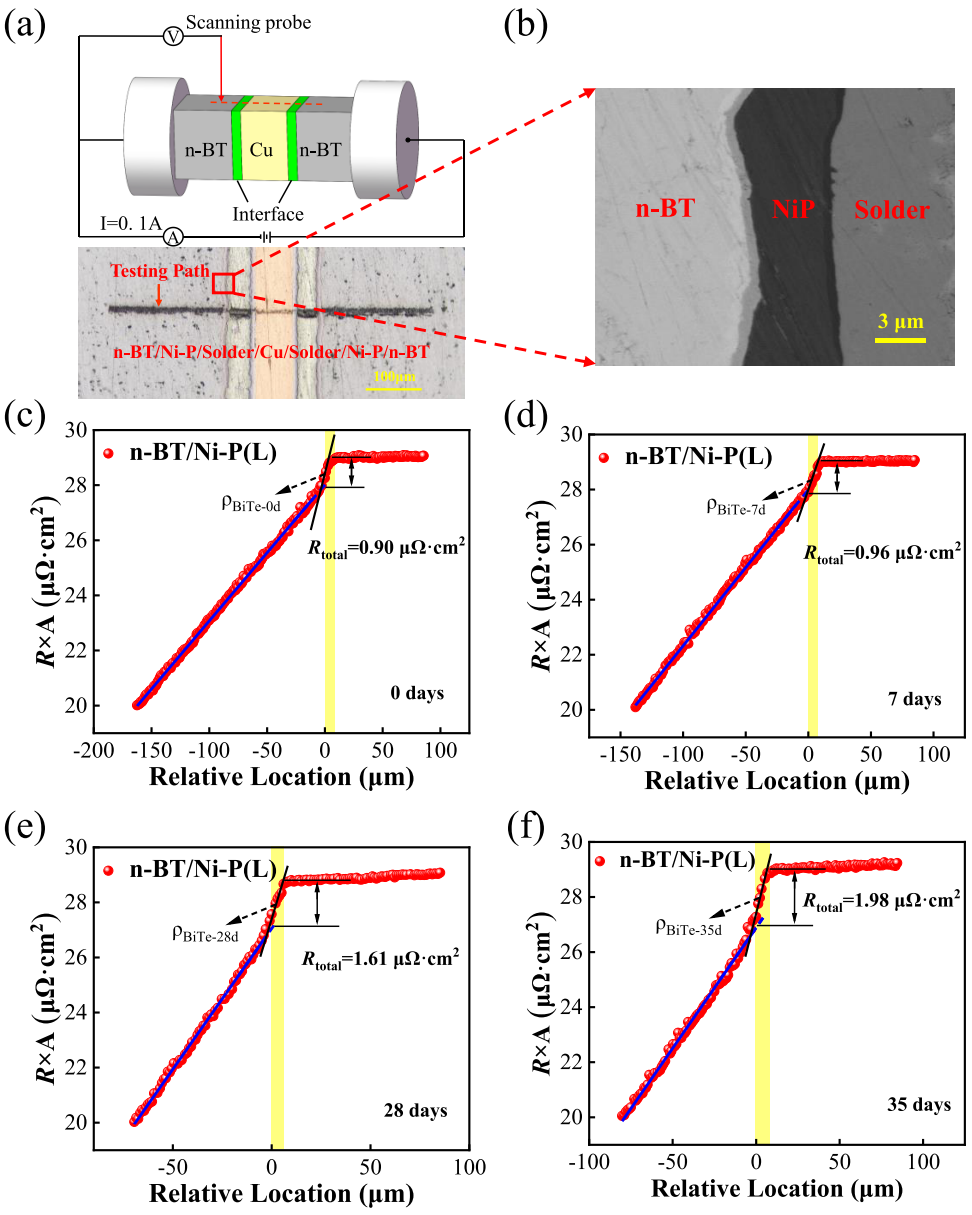


Figure 3. (a) (Top) model diagram of the interface resistance test and (bottom) microscope observation of the image after probe scanning. (b) Localized zoom-in near the interface. The aging of $Bi_2Te_{2.7}Se_{0.3}$/NiP(L) joint at 423 K for (c) 0 days, (d) 7 days, (e) 28 days, and (f) 35 days.

the cross-sectional area of the samples; and L_{NiTe} is the thickness of the NiTe layer at different aging times.

In order to investigate the diffusion kinetics of the NiP barrier, the relationship between IMC thickness and aging time was fitted by the Fick's diffusion laws, namely, $Y_t = Y_0 + (Dt)^{1/2}$, where Y_t, Y_0, D, and t denote the total thickness of dark or bright regions, initial thickness, diffusion rate, and aging time, respectively (Figure 2b). As a result, the diffusion rates of NiP barrier layers are 4.23×10^{-4} $\mu m^2/h$ for NiP(L) and 2.33×10^{-4} $\mu m^2/h$ for NiP(H), respectively, which are much lower than 2.10×10^{-2} $\mu m^2/h$ of pure Ni[32] barrier in n-BT. Therefore, the involvement of P effectively reduces the rate of reaction and diffusion at the interface, due to the blocking effect of P atoms in the NiP lattice and segregated P phase at the interface.

The Sn-based solder has superior solder wettability with NiP(L) barrier layer, while there is a large amount of virtual soldering at NiP(H) so that the high phosphorus content reduces the diffusion rate at the joints, but the solder wettability is poor.[33] The soldered n-type $Bi_2Te_{2.7}Se_{0.3}$ with NiP(L) and NiP(H) was subjected to shear strength test, and the shear strength was measured to be 12.5 ± 0.7 MPa for $Bi_2Te_{2.7}Se_{0.3}$/NiP(L), which is higher than the 4.5 ± 0.9 MPa for $Bi_2Te_{2.7}Se_{0.3}$/NiP(H), indicating the connection reliability of the NiP(L) barrier layer. Therefore, the choice of barrier layers, such as NiP(L), NiP(H), and Ni, can significantly influence the diffusion rates and solder wettability in interfacial connection, consequently affecting the interfacial contact resistivity. These factors should be taken into consideration in the design and fabrication of electronic devices to ensure optimum performance and reliability.

3.2. Interfacial Contact Resistivity in the n-Type $Bi_2Te_{2.7}Se_{0.3}$/NiP(L) Joint. The internal resistance of a thermoelectric device consists of the internal resistance in the n and p semiconductor materials, the electrical resistance of the interface materials, the solder and electrode resistances,

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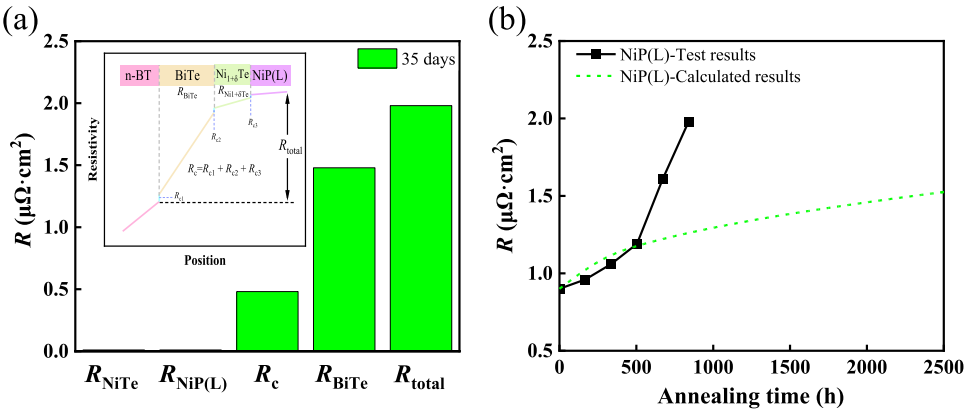


Figure 4. (a) R_{total} denotes the total interfacial contact resistivity after aging at 423 K for 35 days; R_{NiTe}, $R_{NiP(L)}$, and R_{BiTe} denote the contributions of the dark region, interfacial layer, and bright region, respectively; and R_c is the total contact resistivity, with R_{c1} denoting n-BT/BiTe, R_{c2} denoting BiTe/NiTe, and R_{c3} denoting NiTe/NiP(L). (b) Experimental and calculated contact resistivities of NiP(L) layers with different aging times at 423 K.

and the contact resistance of each interface. Among them, the interface contact resistance is affected by the interface layer connection quality and the soldering process. It is known that severe interfacial reactions and elemental diffusion as well as cracks at the interface will lead to a dramatic increase in interfacial contact resistance.[34−36] Consequently, selecting the proper interface material and high-quality interface connections is critical to improving device performance.[37,38]

In order to investigate the long-term stability of the interfacial contact resistivity, the samples with the barrier layer were aged at 423 K for different times, which were applied to measure the n-BT substrate and n-BT/NiP(L) interface resistivity. Then, the measured data was linearly fitted to obtain their accurate resistance. Figure 3a shows the interface resistance test model, where a constant current (I = 0.1 A) is applied to the sample, and the change in voltage is measured by the movement of the probe (step = 1 μm) to obtain the change of interfacial resistance. Figure 3a is the microscope observation of the image after the test, and it can be obviously seen that the testing path of the probe is n-BT/NiP/Solider/Cu/Solder/NiP/n-BT. The interfacial contact resistivity is calculated as $R = \Delta VS/I$ (where, R, ΔV, S, and I are the interfacial contact resistivity, through voltage, sample cross-sectional area, and current). Figure 3b provides a magnified view of the vicinity of the interface.

The initial interfacial contact resistivity of n-BT/NiP(L) is 0.90 $\mu\Omega$ cm^2, which will increase to 0.96, 1.61, and 1.98 $\mu\Omega$ cm^2 when aged at 423 K for 7, 28 and 35 days, respectively, as shown in Figure 3c−f. Therefore, the interfacial contact resistivities of n-BT/NiP(L) joints and the interfacial contact resistivities are gradually becoming larger due to the presence of interfacial reactions and diffusion. But all of these values are much lower than the published results for n-Bi_2Te_3/Ni layers,[17,25,26] revealing the superiority of the NiP barrier layer.

Why the interfacial contact resistivity gradually increased from 0.90 $\mu\Omega$ cm^2 initially to 1.98 $\mu\Omega$ cm^2 with the increase of aging time is worth further analyzing the constitute of contact resistance, which can give a guidance for evaluating the long-term service stability. The total interfacial contact resistance (R_{total}) at the $Bi_2Te_{2.7}Se_{0.3}$/NiP(L) joint consists of additional resistance generated by elemental diffusion to generate BiTe (R_{BiTe}), NiTe compounds generated by interfacial reactions (R_{NiTe}), the bulk resistance of the NiP(L) interface $R_{NiP(L)}$, and the contact resistivity R_c, namely, $R_{total} = R_{BiTe} + R_{NiTe} + R_{NiP(L)} + R_c$. The contact resistivities of BiTe, NiTe, and NiP(L) can all be calculated from the bulk resistivity and thickness. The NiTe compounds generated by the interfacial reaction are semimetallic compounds, which have very low resistivity (~50 $\mu\Omega$ cm), being much lower than $Bi_2Te_{2.7}Se_{0.3}$ (~1000 $\mu\Omega$ cm).[17] Thus, the NiTe generated by the interfacial reaction is not the main reason for the increase in interfacial contact resistivity. The resistivity of BiTe with different aging times is fitted in Figure 3c−f, and the average resistivity of BiTe is calculated to be 1641 $\mu\Omega$ cm, which is much larger than that of the $Bi_2Te_{2.7}Se_{0.3}$ substrate and NiTe IMC layer. Hence, it can be assumed that the BiTe layer has a larger resistivity.

Figure 4a (inset) shows the contributions of BiTe, NiTe, NiP(L), and R_c (the sum of contact resistivities, $R_c = R_{c1} + R_{c2} + R_{c3}$). Figure 4a shows the schematic contact resistivity between the phases of n-BT/BiTe, BiTe/NiTe, and NiTe/NiP(L) to the total interfacial contact resistivity after 35 days of aging at 423 K. It is found that the main contribution to the total interfacial contact resistivity R_{total} comes from the diffusion region (BiTe), which accounts for 74% of the total contact resistivity. Due to the existence of potential barriers for heterogeneous metal contacts, all contact resistivities between n-BT/BiTe, BiTe/NiTe, and NiTe/NiP(L) account for about 20% of the total interfacial resistivity. The interfacial reaction product NiTe and the barrier layer have little effect on the interfacial contact resistivity.

Based on the measured thicknesses of NiTe and as-calculated thicknesses of BiTe, the long-term R_{total} can be predicted according to the diffusion factor obtained in Figure 2b. Figure 4b shows the comparison between the results of interfacial contact resistivity tests and the calculated results; the interfacial contact resistivity increases gradually with the aging time extension, and the calculated results keep the same trend with the actual measured results. The expression for the theoretical contact resistivity is $R = 1.25 \times 10^{-2}\ t^{1/2} + 0.90$ ($\mu\Omega$·cm^2). Since the n-BT/NiP(L) joints are primarily affected by chemical reactions during the preproduction phase, the interfacial contact resistivity grows slowly before aging for 21 days at 423 K within the measurement accuracy range. As aging proceeds, the Te-deficient region for the diffusion is the main reason for the increase in the total interfacial contact resistivity; correspondingly, the contact resistance R_c also increases. And the NiP(L) barrier layer maintains a very low contact resistivity (1.98 $\mu\Omega$ cm^2) after aging at 423 K for 35

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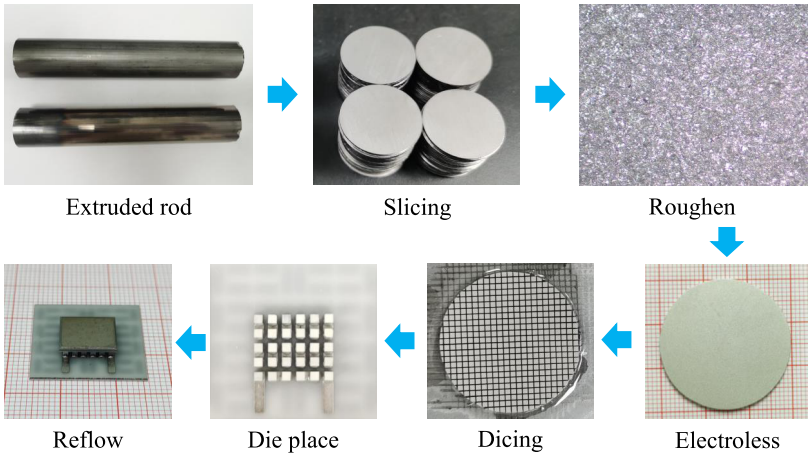


Figure 5. Preparation process of the thermoelectric cooling device.

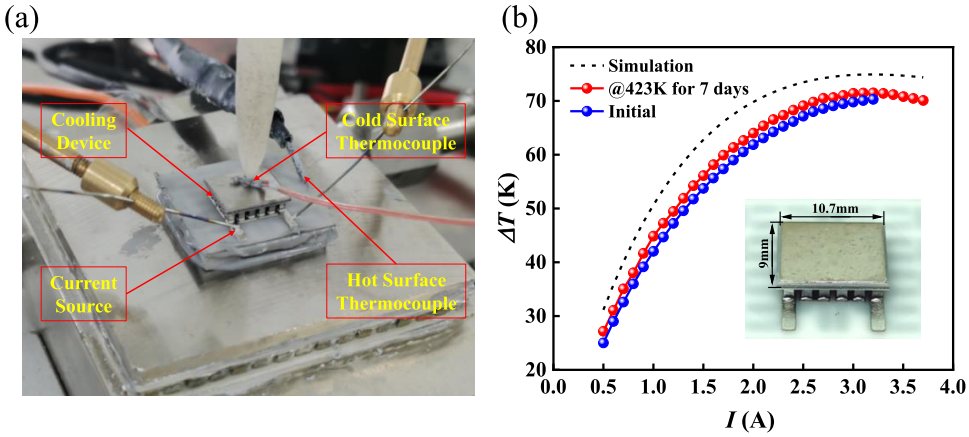


Figure 6. (a) Performance test equipment for TEC. (b) Comparison of experimental and simulated temperature differences for TEC.

days. According to the theoretical equations, the theoretical contact resistivities of n-BT/NiP(L) joints can be predicted to be 2.84 $\mu\Omega$ cm^2 after aging for 1000 days, which exhibits excellent stability compared with the current commercial Ni barrier layer.

3.3. Device Preparation and Performance Measurement. A thermoelectric module usually consists of a metal electrode connecting an n-type thermoelectric material to a p-type thermoelectric material. Due to the low output voltage of a single thermoelectric element, it is often necessary to connect many thermoelectric legs in electrical series and thermal parallel to form a thermoelectric device or assembly for practical application. Figure 5 shows the preparation process of the thermoelectric cooling device. First, commercial n-type and p-type Bi_2Te_3 extruded rods were sliced; second, alumina powder was used to regulate the mechanical roughness to obtain high bond strength; third, the cleaned surface of the sample was chemically plated with NiP(L); fourth, the wafers were scribed and diced into 1 mm × 1 mm × 1.6 mm thermoelectric legs; fifth, according to the distribution of circuits, the placement of n-type and p-type legs; and finally, soldering integrated devices. The initial room-temperature internal resistance of the device is obtained to be 0.5171 Ω, which slightly increased to 0.5513 Ω after aging at 423 K for 7 days.

Figure 6a shows the TEC test equipment. The thermally conductive silicone grease is uniformly coated on the lower substrate of the thermoelectric device to ensure excellent heat conduction, the hot side of the thermoelectric device is heated by a temperature-controlled table, and two T-type thermocouples are placed at the top and the bottom of the device, respectively, in order to measure the temperature of the hot side and the temperature of the cold side. The temperature difference corresponding to different currents of the device is tested by passing current through an external load. Figure 6b shows the simulation and experiment comparison of the TEC current and temperature difference.

When the hot-side temperature is 304 K, the experimentally measured maximum temperature difference of TEC is 70.2 K at a current of 3.2 A. After aging at 423 K for 7 days, the maximum cooling temperature difference of the device was still measured to be 71.5 K, which is slightly lower than the simulated cooling temperature difference of 74.9 K. The presence of thermal radiation during the testing process results in an error of 4.5% in the actual test results and simulation results. The excellent cooling performance is related to the properties of the material itself and also benefits from the good diffusion inhibition, low interfacial resistivity, and thermal stability of the NiP(L) barrier layer.

4. CONCLUSIONS

This paper focuses on investigating the interfacial contact resistivity between the NiP alloy barrier layer and $Bi_2Te_{2.7}Se_{0.3}$ substrate. And the TEC device with high cooling temperature

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difference using the NiP barrier layer was successfully fabricated. The NiP barrier layer effectively reduces diffusion and interfacial reaction compared to the commercial Ni barrier layer, resulting in the low interfacial contact resistivity between $Bi_2Te_{2.7}Se_{0.3}$ and the NiP barrier layer, even aged at 423 K for 35 days. Furthermore, the theoretical interfacial contact resistivity of n-BT/NiP(L) joints as a function of aging time is predicted based on the investigation of the growth rate of the IMC layer. Finally, a 15-pair bismuth telluride device is prepared using the NiP barrier layer, which produces a maximum cooling temperature difference of 71.5 K at a hot-side temperature of 304 K. These results highlight the low interfacial contact resistivity of the NiP barrier layer for n-Bi_2Te_3 and high performance and reliability of Bi_2Te_3-based TEC devices with a NiP barrier layer.

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Author Contributions

R.L. and P.Z. reviewed and revised the manuscript, supervised, and contributed to project management. E.M. carried out writing, experiments, and data processing. Y.L. and J.F. helped with the survey and data organization. L.Z. and Y.X. performed physical phase analysis. L.G. and J.L. provided methodology. All authors analyzed the results.

Notes

The authors declare no competing financial interest.

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REFERENCES

(1) Wu, Z.; Zhang, S.; Liu, Z.; Mu, E.; Hu, Z. Thermoelectric converter: Strategies from materials to device application. *Nano Energy* **2022**, *91*, No. 106692, DOI: 10.1016/j.nanoen.2021.106692.

(2) Chen, W.-Y.; Shi, X.-L.; Zou, J.; Chen, Z.-G. Thermoelectric coolers: Progress, Challenges, and Opportunities. *Small Methods* **2022**, *6*, No. 2101235, DOI: 10.1002/smtd.202101235.

(3) Zhao, Y.; Wang, S.; Ge, M.; Liang, Z.; Liang, Y.; Li, Y. Performance investigation of an intermediate fluid thermoelectric generator for automobile exhaust waste heat recovery. *Appl. Energy* **2019**, *239*, 425−433.

(4) Zhang, Y.; Cleary, M.; Wang, X.; Kempf, N.; Schoensee, L.; Yang, J.; Joshi, G.; Meda, L. High-temperature and high-power-density nanostructured thermoelectric generator for automotive waste heat recovery. *Energy Convers. Manage.* **2015**, *105*, 946−950.

(5) Li, K.; Garrison, G.; Zhu, Y.; Moore, M.; Liu, C.; Hepper, J.; Bandt, L.; Horne, R.; Petty, S. Thermoelectric power generator: Field test at bottle rock geothermal power plant. *J. Power Sources* **2021**, *485*, No. 229266, DOI: 10.1016/j.jpowsour.2020.229266.

(6) Kuroki, T.; Kabeya, K.; Makino, K.; Kajihara, T.; Kaibe, H.; Hachiuma, H.; Matsuno, H.; Fujibayashi, A. Thermoelectric generation using waste heat in steel works. *J. Electron. Mater.* **2014**, *43*, 2405−2410.

(7) Wang, X.; Liang, R.; Fisher, P.; Chan, W.; Xu, J. Critical design features of thermal-based radioisotope generators: A review of the power solution for polar regions and space. *Renewable Sustainable Energy Rev.* **2020**, *119*, No. 109572, DOI: 10.1016/j.rser.2019.109572.

(8) Lorenz, R. D.; Clarke, E. S. Influence of the multi-mission radioisotope thermoelectric generator (MMRTG) on the local atmospheric environment. *Planet. Space Sci.* **2020**, *193*, No. 105075, DOI: 10.1016/j.pss.2020.105075.

(9) O'Brien, R. C.; Ambrosi, R. M.; Bannister, N. P.; Howe, S. D.; Atkinson, H. V. Safe radioisotope thermoelectric generators and heat sources for space applications. *J. Nucl. Mater.* **2008**, *377*, 506−521.

(10) Zhao, L.; Li, H.; Meng, J.; Li, Z. The recent advances in self-powered medical information sensors. *InfoMat* **2020**, *2*, 212−234.

(11) Yuan, C.; Kress, S.; Sadashivaiah, G.; Rudnyi, E. B.; Hohlfeld, D.; Bechtold, T. Towards efficient design optimization of a miniaturized thermoelectric generator for electrically active implants via model order reduction and submodeling technique. *Int. J. Numer. Methods Biomed. Eng.* **2020**, *36*, No. e3311, DOI: 10.1002/cnm.3311.

(12) Wang, Y.; Zhu, W.; Deng, Y.; Fu, B.; Zhu, P.; Yu, Y.; Li, J.; Guo, J. Self-powered wearable pressure sensing system for continuous healthcare monitoring enabled by flexible thin-film thermoelectric generator. *Nano Energy* **2020**, *73*, No. 104773, DOI: 10.1016/j.nanoen.2020.104773.

(13) Hasan, M. N.; Sahlan, S.; Osman, K.; Ali, M. S. M. Energy harvesters for wearable electronics and biomedical devices. *Adv. Mater. Technol.* **2021**, *6*, No. 2000771, DOI: 10.1002/admt.202000771.

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(14) Lin, X.; Mo, S.; Jia, L.; Yang, Z.; Chen, Y.; Cheng, Z. Experimental study and Taguchi analysis on LED cooling by thermoelectric cooler integrated with microchannel heat sink. *Appl. Energy* **2019**, *242*, 232−238.
(15) Söylemez, E.; Alpman, E.; Onat, A. Experimental analysis of hybrid household refrigerators including thermoelectric and vapour compression cooling systems. *Int. J. Refrig.* **2018**, *95*, 93−107.
(16) Xiong, K.; Wang, W. C.; Alshareef, H. N.; Gupta, R. P.; White, J. B.; Gnade, B. E.; Cho, K. J. Electronic structures and stability of Ni/Bi_2Te_3 and Co/Bi_2Te_3 interfaces. *J. Phys. D: Appl. Phys.* **2010**, *43*, No. 115303, DOI: 10.1088/0022-3727/43/11/115303.
(17) Liu, W. S.; Wang, H. Z.; Wang, L. J.; Wang, X. W.; Joshi, G.; Chen, G.; Ren, Z. F. Understanding of the contact of nanostructured thermoelectric n-type $Bi_2Te_{2.7}Se_{0.3}$ legs for power generation applications. *J. Mater. Chem. A* **2013**, *1*, 13093−13100.
(18) Chen, S.-W.; Chiu, C.-N. Unusual cruciform pattern interfacial reactions in Sn/Te couples. *Scr. Mater.* **2007**, *56*, 97−99.
(19) Zhang, H.; Jing, H. Y.; Han, Y. D.; Xu, L. Y.; Lu, G. Q. Interfacial reaction between n- and p-type thermoelectric materials and SAC305 solders. *J. Alloys Compd.* **2013**, *576*, 424−431.
(20) Chen, S.-W.; Wu, H.-J.; Wu, C.-Y.; Chang, C.-F.; Chen, C.-Y. Reaction evolution and alternating layer formation in Sn/$(Bi_{0.25}Sb_{0.75})_2Te_3$ and Sn/Sb_2Te_3 couples. *J. Alloys Compd.* **2013**, *553*, 106−112.
(21) Liao, C.-N.; Chen, W. T.; Lee, C. H. Polarity effect on interfacial reactions at soldered junctions of electrically stressed thermoelectric modules. *Appl. Phys. Lett.* **2010**, *97*, No. 241906, DOI: 10.1063/1.3525930.
(22) Lo, L.-C.; Wu, A. T. Interfacial reactions between diffusion barriers and thermoelectric materials under current stressing. *J. Electron. Mater.* **2012**, *41*, 3325−3330.
(23) Lan, Y. C.; Wang, D. Z.; Chen, G.; Ren, Z. F. Diffusion of nickel and tin in p-type $(Bi,Sb)_2Te_3$ and n-type $Bi_2(Te,Se)_3$ thermoelectric materials. *Appl. Phys. Lett.* **2008**, *92*, No. 101910, DOI: 10.1063/1.2896310.
(24) Chen, S.-W.; Yang, T.-R.; Wu, C.-Y.; Hsiao, H.-W.; Chu, H.-S.; Huang, J.-D.; Liou, T.-W. Interfacial reactions in the Ni/$(Bi_{0.25}Sb_{0.75})_2Te_3$ and Ni/$Bi_2(Te_{0.9}Se_{0.1})_3$ couples. *J. Alloys Compd.* **2016**, *686*, 847−853.
(25) Tang, H.; Bai, H.; Yang, X.; Cao, Y.; Tang, K. C.; Zhang, Z. K.; Chen, S.; Yang, D. W.; Su, X. L.; Yan, Y. G.; Tang, X. F. Thermal stability and interfacial structure evolution of Bi_2Te_3-based micro thermoelectric devices. *J. Alloys Compd.* **2022**, *896*, No. 163090, DOI: 10.1016/j.jallcom.2021.163090.
(26) Joshi, G.; Mitchell, D.; Ruedin, J.; Hoover, K.; Guzman, R.; Mcaleer, M.; Wood, L.; Savoy, S. Pulsed-light surface annealing for low contact resistance interfaces between metal electrodes and bismuth telluride thermoelectric materials. *J. Mater. Chem. C* **2019**, *7*, 479−483.
(27) Liu, R. H.; Chen, H. Y.; Zhao, K. P.; Qin, Y. T.; Jiang, B. B.; Zhang, T. S.; Sha, G.; Shi, X.; Uher, C.; Zhang, W. Q.; Chen, L. D. Entropy as a gene-like performance indicator promoting thermoelectric materials. *Adv. Mater.* **2017**, *29*, No. 1702712, DOI: 10.1002/adma.201702712.
(28) Zhu, W.; Wei, P.; Zhang, J. Q.; Li, L. Z.; Zhu, W. T.; Nie, X. L.; Sang, X. H.; Zhang, Q. J.; Zhao, W. Y. Fabrication and excellent performances of bismuth telluride-based thermoelectric devices. *ACS Appl. Mater. Interfaces* **2022**, *14*, 12276−12283.
(29) Wang, C.-H.; Hsieh, H.-C.; Sun, Z.-W.; Ranganayakulu, V. K.; Lan, T.-W.; Chen, Y.-Y.; Chang, Y.-Y.; Wu, A. T. Interfacial stability in Bi_2Te_3 thermoelectric joints. *ACS Appl. Mater. Interfaces* **2020**, *12*, 27001−27009.
(30) Lin, W.-C.; Li, Y.-S.; Wu, A. T. Study of diffusion barrier for solder/n-type Bi_2Te_3 and bonding strength for p- and n-type thermoelectric modules. *J. Electron. Mater.* **2018**, *47*, 148−154.
(31) Lin, T. Y.; Liao, C. N.; Wu, A. T. Evaluation of diffusion barrier between lead-free solder systems and thermoelectric materials. *J. Electron. Mater.* **2012**, *41*, 153−158.
(32) Chen, S. W.; Liu, Z. W.; Chu, H. S.; Huang, Z. Y. Interfacial reactions between Ni and $Bi_2(Se_{0.1}Te_{0.9})_3$ and its constituent material systems. *J. Alloys Compd.* **2018**, *731*, 111−117.
(33) Chow, Y. M.; Lau, W. M.; Karim, Z. S. Surface properties and solderability behaviour of nickel-phosphorus and nickel-boron deposited by electroless plating. *Surf. Interface Anal.* **2001**, *31*, 321−327.
(34) Hatzikraniotis, E.; Zorbas, K. T.; Samaras, I.; Kyratsi, T.; Paraskevopoulos, K. M. Efficiency study of a commercial thermoelectric power generator (TEG) under thermal cycling. *J. Electron. Mater.* **2010**, *39*, 2112−2116.
(35) Guo, J.; Fan, W.; Wang, Y.; Wang, Y.; Chen, J.; Jiang, Y.; Chen, S. Study on improving comprehensive property of Te-based thermoelectric joint. *J. Alloys Compd.* **2021**, *886*, No. 161242, DOI: 10.1016/j.jallcom.2021.161242.
(36) Wu, W.; Ren, G.-K.; Chen, X.; Liu, Y.; Zhou, Z.; Song, J.; Shi, Y.; Jiang, J.-M.; Lin, Y.-H. Interfacial advances yielding high efficiencies for thermoelectric devices. *J. Mater. Chem. A* **2021**, *9*, 3209−3230.
(37) Chu, J.; Huang, J.; Liu, R.; Liao, J.; Xia, X.; Zhang, Q.; Wang, C.; Gu, M.; Bai, S.; Shi, X.; Chen, L. Electrode interface optimization advances conversion efficiency and stability of thermoelectric devices. *Nat. Commun.* **2020**, *11*, No. 2723, DOI: 10.1038/s41467-020-16508-x.
(38) Gu, M.; Bai, S. Q.; Wu, J. H.; Liao, J. C.; Xia, X. G.; Liu, R. H.; Chen, L. D. A high-throughput strategy to screen interfacial diffusion barrier materials for thermoelectric modules. *J. Mater. Res.* **2019**, *34*, 1179−1187.

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